Engineering review of the Band 3 pumping-station PFD

N-19082-RO-012-TQT-AES-PFD-B3-001 to005, Rev0

Review of all five supplied Rev0 PFD sheets and the two engineer comments in the user message, dated 13 October 2026 (Dubai). Evidence is the visible drawing and extracted labels, checked at full-page and enlarged detail scales. The referenced TAQA specifications/P&IDs, vendor data and hydraulic/transient calculations were not supplied. Not shown means absent from this PFD, not proof that the eventual installation or vendor package omits the device. Major flow paths, interface functions and isolation topology should be resolved on the PFD; detailed valve types, sizes, classes, settings and minor vents/drains require P&ID/vendor confirmation. High means resolve the operating/safety/availability function before design closeout; Medium means clarify or close through detailed design; Low means drawing consistency. This is a review record, not an approved design or a revised process drawing.

Disposition: revise the identified functional arrangements and reconcile with P&IDs before technical closeout.

The principal concerns are pump suction isolation, independently isolable surge trains, discharge-header availability, chemical dosing across different pressures, potable backflow protection, compatible chemical drainage and controlled lagoon release.

Both supplied engineer comments remain open. WS-500 lacks the carbon-filter designation, and TK-504 has no cooling interface.

What is already present

- Normal water path is coherent: incoming pipelines -> storage -> three suction sections -> separate transmission/distribution pump groups -> separate external outlets. Gravity supplies pump suction; the main pumps provide the onward pressure.

- Reference flow arithmetic is consistent: 200 MIGD = 37,884.083 m3/h; six 25 MIGD duty pumps give 150 MIGD transmission and two give 50 MIGD distribution. One standby and one future bay are shown per current group.

- Individual tank filling/outlet MOVs, normal-open sectioning valves and the normally closed station bypass are present.

- All ten current main-pump discharge positions show an NRV and a downstream isolation valve. Their orientation is consistent with the legend; the filled check-valve half must not be mistaken for a separate wrong-way flow arrow.

- Metering, injection, sampling/analyser and outlet FCV/MOV functions are included. Injection follows metering and precedes sampling.

- Suction bladder surge vessels are correctly separated from the air-compressor supply; discharge vessels have an air-main interface.

- Duty/standby oil-free air compressors with aftercooler/separator and dryer/filter trains, two receivers and a mobile charging connection are included.

- Pump/motor, VFD and air-compressor cooling are deliberately separate. The closed C1 loop returns to HX-701, C2 has supply/return, and C3 is air cooled with no water connection. Electrolyzer CH-501 is a separate package duty.

- Clean/general, hyperchlorination and chemical containment destinations are differentiated conceptually; treatment and compatible chemical segregation still require closure.

- The P-321 future suction crosses the existing distribution discharge without connection, using the drawing gap convention. The principal reciprocal off-sheet service/destination references can be traced across002-005.

Sheet-by-sheet comments

R01 | Medium | Sheet 001 / 002 | Design basis / storage

Location: T-101 to T-108; 24 h storage statement

Observed: Eight tanks labelled 25 MIG give 200 MIG, exactly 24 h at 200 MIGD. The drawing does not identify whether 25 MIG is usable or gross volume.

Required action: Define gross and usable storage, dead volume, minimum operating level and any reserved volume. If 25 MIG is gross, usable autonomy is less than 24 h. With one nominal 25 MIG tank unavailable, gross autonomy is 21 h; the drawing does not claim 24 h N-1, so confirm whether that is required.

Close using: Storage calculation / design basis / PFD note | Calculation required

R02 | High | Sheet 002 | Station inlet

Location: Three typical incoming lines upstream of the common inlet manifold

Observed: Each incoming line shows a MOV, FCV and FE/FIT, but no line check valve or stated reverse-flow strategy.

Required action: Confirm actual incoming line count and network pressure combinations. Establish how backfeeding between live incoming pipelines is prevented or intentionally permitted; show check/isolation arrangements only as required by the network authority and hydraulic study. Do not assume every inlet needs an NRV.

Close using: Network interface / hydraulic study / P&ID | Confirmation required

R03 | Medium | Sheet 002 | Station inlet

Location: MOV-101 / FCV-101 / FE-101 typical inlet control

Observed: Upstream motorized isolation is shown. Local downstream isolation, FCV maintenance bypass and metering maintenance provisions are not shown.

Required action: Confirm the maintenance philosophy. Where one inlet control train must remain maintainable with the other lines live, provide suitable isolation on both sides of the equipment. A bypass is conditional, not automatically required. Check meter straight lengths, FCV cavitation/noise and valve fail/closure behavior in the P&ID and calculations.

Close using: P&ID / FCV and flowmeter data | P&ID confirmation

R04 | High | Sheet 002 / 003 | Storage / pump protection

Location: LIT-101 (TYP.), tank inlet MOVs and main pumps

Observed: A typical level transmitter and tank inlet/outlet MOVs are shown, but the functional high-high filling shutdown and low-low pump protection are not stated.

Required action: Add the operating philosophy: tank high-high level inhibits/closes the appropriate filling path; tank/header low level or low suction pressure prevents unsafe pump operation. Define valve proof, pump sequencing and independent protection requirements in the P&ID/cause-and-effect. Do not invent trip settings.

Close using: PFD note / P&ID / cause-and-effect | Review finding

R05 | High | Sheet 002 / 005 | Tank drainage

Location: Each tank overflow and bottom drain to OP-101

Observed: Side overflow and bottom drain connections join dashed lines to OP-101. Individual bottom-drain isolation is not shown.

Required action: Distinguish the normally closed, isolable bottom drain from the overflow, which must remain available. Document screened/sanitary termination, air gap or equivalent backflow protection, venting and capacity. Prevent lagoon

or pit water from reaching potable storage. Do not place an ordinary shutoff valve in the required overflow path.

Close using: PFD clarification / tank and drainage P&IDs | Review finding

R06 | High | Sheet 002 / 003 | Station bypass

Location: MOV-102 (NC), inlet manifold to suction header

Observed: The inlet-to-suction bypass and its normally closed MOV are present. The pressure transition and mode interlocks are not described.

Required action: Define bypass mode using actual incoming pressure. Check pressure exposure of the suction system and tank outlet lines, potential reverse filling through tank outlets, NPSH, surge and isolation proof. Determine required pressure reduction, backflow protection and tank-isolation interlocks from the approved arrangement.

Close using: Hydraulic study / PFD operating note / P&ID | Confirmation required

R07 | High | Sheet 002 / 003 | Gravity feed

Location: Tank outlet manifold to three suction-header sections

Observed: The normal path is correctly shown as storage to suction header to main pumps. No water levels or hydraulic-loss data establish the gravity-feed performance.

Required action: Verify the hydraulic grade and NPSH available at minimum tank level for both groups, with the intended maximum simultaneous flow and unavailable tanks/sections. Include entrance, manifold, suction-valve and pipe losses; compare against actual pump NPSH requirements and project margin.

Close using: Hydraulic / NPSH calculation | Calculation required

R08 | High | Sheet 003 | Main pumps

Location: P-301 to P-307 and P-311 to P-313 suction branches

Observed: Discharge NRVs and isolation valves are drawn for all ten current main-pump positions. Individual suction isolation valves are not drawn.

Required action: Show one suitable suction isolation provision for each current main pump, with future-pump provisions separately identified. Retain the existing discharge sequence pump -> NRV -> isolation -> header. Add pump casing drain/vent and local pressure measurement on the P&ID; confirm valve type and hydraulic impact.

Close using: PFD / main-pump P&ID | Review finding

R09 | High | Sheet 003 | Header resilience

Location: Group A and Group B sectioning valves; Note 2

Observed: Each discharge header is sectioned, but each has one outlet on the right side. A left-hand section cannot supply that outlet when the intervening sectioning valve is closed.

Required action: Demonstrate the claim that at least half of each group remains available with any one section isolated. Provide a suitable independently isolable outlet/header arrangement, or narrow the note to the actual supported isolation cases. Check suction and discharge sections together; pump count alone does not prove an available outlet path.

Close using: PFD topology / isolation-case hydraulic matrix | Review finding

R10 | Medium | Sheet 003 | Future equipment

Location: P-308, P-314 and future second distribution group P-321

Observed: Future pump bays and a future group with separate discharge/surge provision are shown. P-321 is fed from the suction header and crosses S-22 without connection, consistent with the drawing gap convention.

Required action: Identify installed versus future valves/spools, capped or blinded live tie-ins, and the future duty/standby arrangement. Preserve the separation of the future suction connection from the existing distribution discharge header.

Close using: PFD / tie-in schedule / P&ID | Clarification required

R11 | Medium | Sheet 003 | Group outlets

Location: FCV-401 / FCV-411 and their bypasses

Observed: Both outlet FCVs have a manual-valve bypass and a downstream common MOV. Separate block valves on the two ends of the FCV leg are not drawn; bypass normal status is unstated.

Required action: If live FCV maintenance is intended, add isolation at each end of the FCV leg, leaving the bypass path available. The downstream common MOV alone cannot isolate the FCV while bypass flow continues. State NC/secured bypass status and approved manual operation; otherwise document whole-outlet shutdown maintenance.

Close using: PFD normal status / outlet P&ID | Review finding

R12 | High | Sheet 003 | Pump / outlet control

Location: VFD pumps, outlet FCVs/MOVs and main headers

Observed: VFDs and flow instrumentation are present, but main suction/discharge pressure measurement and the coordinated pressure/flow operating philosophy are not shown.

Required action: Define how VFD control and outlet FCV control interact, minimum safe pump flow and valve/pump start-stop sequencing. Confirm pressure measurement, low suction/no-flow protection and MOV closure times in the P&ID/cause-and-effect/transient study. The stated 250 m and 100-160 m heads are differential heads, not complete valve or vessel design pressures.

Close using: PFD functional note / P&ID / surge study | Review finding

R13 | Medium | Sheet 003 | Water quality

Location: IP-401 / SP-401 / WQM-401 and IP-411 / SP-411 / WQM-411

Observed: Metering upstream of injection and sampling/analyser points downstream of injection are correctly sequenced. Sufficient chemical mixing and representative sample conditions are not established.

Required action: Confirm mixing length or mixer requirements and sample-system design for the actual analytes, especially residual chlorine. Add analyser sample isolation, conditioning and safe drain details in the P&ID. Do not infer adequate contact or mixing from schematic spacing.

Close using: Mixing calculation / analyser data / P&ID | Confirmation required

R14 | Medium | Sheet 003 | Suction surge vessels

Location: SV-201 / SV-202 / SV-203 typical pairs

Observed: Each suction section has bladder vessels behind one common water-side isolation valve. The note correctly excludes them from the compressed-air system.

Required action: Confirm individual-vessel or train maintenance isolation, safe drain/vent and manufacturer-approved precharge provisions. Prove sufficient surge protection for permitted out-of-service cases. Keep the required bidirectional hydraulic response; do not add an ordinary NRV to the vessel water connection without a deliberately engineered surge arrangement.

Close using: Surge study / vessel vendor P&ID | Review finding

R15 | High | Sheet 003 | Discharge surge trains

Location: SV-401 and SV-411, Train 1 and Train 2

Observed: For each group, one takeoff valve feeds Train 1, followed by a second water-line valve between Train 1 and Train 2. This isolates Train 2 alone, or both trains; it cannot isolate Train 1 while retaining Train 2.

Required action: Provide independently isolable parallel train branches, or another arrangement that demonstrably meets the required maintenance cases. Show protected normal-open status, independent air-side isolation/control, drains/vents and the remaining active surge capacity. Revalidate the arrangement by transient analysis.

Close using: PFD topology / surge study / vendor P&ID | Review finding

R16 | High | Sheet 003 / 005 | Surge compressed air

Location: Common S-14 air main to transmission and distribution surge vessels

Observed: The same common air main supplies both groups, despite their different pump head bands. Per-group or per-vessel air pressure/level controls and water-backflow safeguards are not shown.

Required action: Establish actual vessel operating/charge pressures and provide the required regulated/control branches, isolation, NRVs or equivalent water-backflow safeguards and pressure/level protection. A 250 m water head is about 24.52 bar differential; 100-160 m is about 9.81-15.69 bar. Add suction/static pressure and transient limits before selecting compressor or receiver ratings; these conversions are not setpoints.

Close using: Surge / compressor sizing / vendor P&ID | Confirmation required

R17 | High | Sheet 005 | Compressor packages

Location: C-601 / C-602, DR-601 / DR-602 to common receiver feed

Observed: Oil-free duty/standby compressors, intake equipment, aftercooler/separator and dryer/filter trains are present. Separate discharge NRVs and maintenance isolation are not drawn before the trains join.

Required action: Confirm these devices within the package and show/reference the outlet isolation and non-return function so a live receiver cannot backfeed the standby/open compressor train. Verify separator/aftercooler/dryer drains and discharge destinations. Confirm air quality and dew point against the actual W-M-SS-007 requirements.

Close using: Compressor package P&ID / project specification | P&ID/OEM confirmation

R18 | High | Sheet 005 | Air receivers

Location: V-601 / V-602

Observed: Two receivers are connected to common upstream/downstream air lines. Individual isolation, pressure indication, safety relief and low-point condensate drains are not drawn.

Required action: Confirm individual service isolation and receiver safety relief/pressure indication/drainage in the vendor P&ID. Relief protection must remain effective for every permitted isolation state; ordinary maintenance valves must not defeat required vessel protection. Establish working/design pressure from the surge duty.

Close using: Receiver data / vendor P&ID | P&ID/OEM confirmation

R19 | Medium | Sheet 005 | Mobile compressor connection

Location: MC-601

Observed: A manual valve and capped-looking connection are drawn at the common air main. Backflow prevention and the mobile source pressure/quality limits are not stated.

Required action: Confirm a positively capped/blinded connection, isolation and suitable NRV/backflow safeguard, with approved pressure regulation and connection procedure. The mobile source must meet the required pressure and air quality; prevent uncontrolled charging or discharge toward disconnected equipment.

Close using: P&ID / charging procedure | P&ID confirmation

R20 | Medium | Sheet 005 | Service water

Location: P-801 / P-802 and utility branches

Observed: Duty/standby pumps each have a discharge NRV. Individual suction/discharge isolation and consumer branch isolation are not drawn.

Required action: Confirm service-pump maintenance isolation, pressure control/protection and isolation for washdown, domestic water, disinfection and cooling makeup. Retain the two existing pump NRVs. Confirm independent domestic/potable and process-service classifications with the water authority.

Close using: Service-water P&ID | P&ID confirmation

R21 | High | Sheet 004 / 005 | Potable backflow protection

Location: S-41 to WS-500 and service-water supply to P-703

Observed: Potable-derived service water supplies the salt/chemical process and an inhibitor-containing cooling loop; hazard-appropriate backflow protection is not shown.

Required action: Provide an approved separation/backflow arrangement at the chemical pretreatment/process supply and cooling-makeup interface, and at other hazardous utility consumers. Select air gap or appropriate tested devices according to the actual hazard and authority requirements; a pump NRV alone is not complete contamination protection.

Close using: PFD interface note / water authority / P&ID | Review finding

R22 | High | Sheet 005 | Pump / motor cooling

Location: HX-701 / P-702 / HX-707 closed loop

Observed: A closed cooling circuit, heat-rejection interface, main pump/motor coolers and a parallel lube-oil HX are shown. Equipment quantities, heat rejection type, duties and loss-of-cooling behavior are unstated.

Required action: Define the actual primary/secondary heat-rejection arrangement and ambient design conditions, pump/motor/lube-oil heat loads, supply/return temperatures, flow and redundancy. Confirm cooling proof and temperature/low-flow protection required for the main pumps. The two HX circuits must remain hydraulically distinct where indirect exchange is intended.

Close using: PFD equipment duties / OEM thermal design / P&ID | Calculation/OEM confirmation

R23 | Medium | Sheet 005 | Cooling piping

Location: P-702 / P-703; F-704 / V-705 / X-706; cooler branches

Observed: Filters, expansion/pressure vessel and inhibitor dosing are combined in one box connected by a single branch. Component piping roles and maintenance/balancing valves are not shown.

Required action: Clarify whether filtration is full-flow or side-stream; show the expansion connection and dosing injection separately. Confirm circulation/makeup pump NRVs where needed, isolations, cooler branch isolation/balancing, vents, drains and pressure/thermal relief. Define the allowable condition when an expansion-vessel connection is isolated.

Close using: Cooling P&ID / package diagram | P&ID confirmation

R24 | High | Sheet 005 | VFD cooling

Location: C2 drive-OEM package and VFD units

Observed: Supply and return are shown between a generic VFD cooling package and the drives. The ultimate heat sink, heat load and failure response are not identified.

Required action: State whether the OEM system rejects heat to ambient air, a dedicated chiller or another separately defined circuit. Add its required external interfaces and thermal duty; confirm flow/temperature monitoring, maintenance isolation and drive derating/trip behavior. Do not connect it to C1 or CH-501 merely to close the drawing.

Close using: PFD interface / VFD OEM cooling design | OEM confirmation

R25 | Medium | Sheet 004 | Electrolyzer pretreatment

Location: WS-500

Observed: WS-500 is labelled SOFTENER SKID only. The engineer requested carbon filter / softener skid.

Required action: Revise the label to CARBON FILTER / SOFTENER SKID and identify the actual process arrangement and package boundary. Confirm treated-water chlorine/hardness quality and consumables/regeneration against the electrolyzer vendor; the revised label must match the actual supplied equipment.

Close using: PFD / vendor pretreatment package | Engineer comment 1 - not closed

R26 | Medium | Sheet 004 | NaOCl storage cooling

Location: TK-504 and CH-501; Sheet 004 Note 3

Observed: No chilled-water supply/return or storage cooling interface is shown at TK-504. CH-501 and Note 3 explicitly identify softener/electrolyzer duties only.

Required action: Add a clearly marked chilled-water supply/return or dedicated storage-cooling interface/note for TK-504 as requested. Update CH-501 duty description and Note 3 if it also serves storage. Select an OEM-approved indirect cooling method (jacket/coil/external HX as applicable), rather than connecting chilled water into the chemical contents. Confirm storage temperature, compatible materials and cooling duty.

Close using: PFD / thermal duty / vendor storage design | Engineer comment 2 - not closed

R27 | Medium | Sheet 004 | Electrolyzer feed / chilled water

Location: CH-501 drawn in the softened-water feed route

Observed: The route from WS-500 passes through the CH-501 symbol to EL-503 as one process line. No separate chilled-water circuit is drawn.

Required action: Clarify whether this is cooled softened process water through an integrated feed-water chiller or an indirect chilled-water HX. Identify the cooled process stream and external CHWS/CHWR interfaces separately, with the required circulation/temperature control. A chilled-water plant label alone does not define the fluid circuits.

Close using: PFD process definition / vendor P&ID | Clarification required

R28 | High | Sheet 003 / 004 | Hypochlorite dosing

Location: P-506 split to S-16 / IP-401 and S-17 / IP-411

Observed: One typical dosing pump discharges to two branches without drawn independent metering or injection NRV/isolation. The receiving headers operate at different head bands.

Required action: Show independent controllable dosing duties for each header: separate dosing trains or a proven common skid with independent branch metering/control. Confirm injection check/isolation, pressure capability, required backpressure devices and dosing-pump relief to a compatible contained destination. Prevent process water reaching TK-504 or transferring between headers through chemical piping. Base dosage on each measured water flow and approved residual-control philosophy.

Close using: PFD dosing arrangement / dosing calculations / P&ID | Review finding

R29 | High | Sheet 004 | Hydrogen ventilation

Location: EL-503 / TK-504 / B-505 hydrogen vent

Observed: A tank hydrogen vent and dilution-air blower are correctly shown. Ventilation proof, hydrogen handling at the electrolyzers and loss-of-dilution response are not defined.

Required action: Confirm the OEM hydrogen generation, dilution and vent route for both cells and storage; verify safe termination and gas accumulation avoidance. Define blower duty/standby requirements, ventilation proof/interlock and hydrogen detection where required by the actual hazard assessment. No hydrogen limits, air rates or area classification can be selected from this PFD alone.

Close using: PFD functional note / OEM design / hazard assessment | Review finding

R30 | High | Sheet 004 / 005 | Chemical containment

Location: TK-505 HCl / TK-504 NaOCl / UB-501 / CS-821

Observed: Bund drainage is represented as one stream to one chemical containment item. Separate acid and hypochlorite spill paths are not demonstrated.

Required action: Show physically segregated HCl and hypochlorite containment/drain paths or a specifically engineered compatible arrangement. Mixing acid and hypochlorite can release chlorine gas. State normally closed controlled bund-drain operation and identify sampling/collection/disposal; do not route both uncontrolled spill streams to one common sump.

Close using: PFD segregation / chemical drainage P&ID / containment design | Review finding

R31 | Medium | Sheet 004 / 005 | Package effluents

Location: WS-500 / TK-501 / TK-505 / EL-503

Observed: Carbon/softener backwash or regeneration waste, brine/process reject and acid-cleaning effluent destinations are not identified.

Required action: Identify the actual waste streams, quantities and compatible containment/treatment destinations. Show the cleaning transfer/flush function and containment of acid-cleaning effluent. Distinguish brine, acid, hypochlorite and ordinary washdown; the exact package valving belongs in its P&ID.

Close using: PFD waste interfaces / vendor P&ID / treatment design | Confirmation required

R32 | High | Sheet 005 | Pump-hall sump

Location: P-811 / P-812 discharge to TK-831

Observed: Two submersible pumps join a common discharge without drawn individual NRVs or discharge isolation.

Required action: Confirm an individual discharge NRV and service isolation for each sump pump so the duty pump cannot send water through the standby pump into the pit. Add level-based duty/standby controls, high-high alarm and usable sump/holding volumes in the P&ID and drainage calculations.

Close using: Drainage P&ID / sump sizing | Review finding

R33 | High | Sheet 003 / 005 | Header drainage

Location: Typical header drain valve to TK-831

Observed: One typical drain takeoff is shown on the Group B discharge, with a connection to the drainage holding tank. Complete isolated-section drains and controlled depressurization are not demonstrated.

Required action: Provide the necessary low-point drains and high-point venting for suction/discharge and isolated sections, based on the actual pipe profile. Establish safe staged pressure letdown, receiving-tank venting and energy dissipation for high-pressure discharge-header drainage. Avoid an uncontrolled pressurized connection into a civil holding

tank.

Close using: PFD drain note / P&ID / depressurization calculation | Review finding

R34 | High | Sheet 004 / 005 | NaOCl product drainage

Location: TK-504 bottom drain to LG-102 labelled HYPERCHLORINATION DRAINAGE

Observed: The chemical storage tank bottom drain goes directly to the hyperchlorination-water lagoon route. Product-drain concentration and lagoon suitability are not stated.

Required action: Distinguish stock hypochlorite product drains from chlorinated pipeline/disinfection wash water. Route concentrated product to a compatible contained collection/treatment system, or explicitly prove LG-102 is designed for that duty. Identify concentration, volume, controlled drain procedure and treatment before any release.

Close using: PFD destination / chemical-drainage and treatment design | Review finding

R35 | High | Sheet 005 | Lagoon transfer / discharge

Location: LG-102 and LG-103 transfer to LG-101; storm-water connection

Observed: Both drain lagoons connect to a transfer route to LG-101, which has an overflow/bypass to storm water. No controlled release, dechlorination or water-quality acceptance function is shown.

Required action: Define gravity/pumped transfer and reverse-flow prevention from actual levels. Hold and dechlorinate hyperchlorination effluent to the approved discharge criterion before transfer/release. Confirm contaminated pump-hall water acceptance/treatment, transfer isolation and monitoring, lagoon/overflow capacity and authority approval for the storm-water interface.

Close using: PFD treatment/release function / drainage P&ID / discharge approval | Review finding

R36 | Medium | Sheet 005 | Pump-hall drainage quality

Location: Pump/motor/lube-oil areas to PT-811 / LG-103

Observed: Lube-oil heat-exchange equipment is shown and floor/pump/valve drainage joins LG-103. Treatment or separate containment of possible oil-contaminated water is unstated.

Required action: Confirm whether the actual pump/lube equipment can introduce oil into floor drains. If so, provide suitable containment/separation and controlled disposal before clean drainage/storm-water release; otherwise record the clean-drain basis. Do not assume every pump-hall drain automatically requires an oil separator.

Close using: Drainage classification / OEM leakage data / treatment design | Confirmation required

R37 | Medium | Sheet 001-005 | Review scope / traceability

Location: Referenced W-G-SS-017, W-M-SS-007 and N14054 typical P&IDs

Observed: The cited specifications, reference P&IDs, vendor package P&IDs and detailed hydraulic/transient studies were not supplied with this five-sheet PDF.

Required action: Use the reviewed PFD as a functional baseline, then reconcile it with the actual project and package P&IDs, valve schedules and approved studies. This review cannot establish that every installed valve exists or verify the cited standard clauses, sizes, pressure classes or settings.

Close using: Referenced specifications / P&IDs / vendor data | Review limitation

R38 | Low | Sheet 001 / 002 / 003 | Standard drawing clarity

Location: Band-3 scalability table and functional/TYP tags

Observed: The Band-3 example table includes 100 MIGD, while the stated band definition places 100 MIGD in Band 2. Several repeated valves/vessels use a single functional or typical tag.

Required action: Label 100 MIGD as a boundary example or align the table with the band definition. Retain functional/TYP identification at PFD level if intended, but assign unique equipment and valve identities when the project P&IDs/schedules are issued. Do not count each typical symbol as a certified bill of materials.

Close using: PFD editorial clarification / project tag schedule | Drawing clarity

Valve checklist

This is a visible-symbol/function checklist, not a complete valve bill of materials. Current main-pump positions exclude future pump bays; typical quantities are illustrative.

Sheet 002 | Incoming supply control

Shown / expected: 3 typical lines: MOV + FCV each | Status: Present

Check inlet isolation/backflow strategy and live maintenance; R02-R03

Sheet 002 | Tank inlet/outlet

Shown / expected: 8 inlet MOVs + 8 outlet MOVs | Status: Present

Level control and proof; R04. Functional tags require project identities.

Sheet 002 | Tank outlet sectioning

Shown / expected: 2 manual section valves marked NO | Status: Present

Check gravity flow and isolation cases; R07/R09

Sheet 002 / 003 | Station bypass

Shown / expected: MOV-102 marked NC | Status: Present

Pressure suitability and bypass-mode interlocks; R06

Sheet 002 | Service-water takeoff

Shown / expected: 1 manual branch isolation | Status: Present

Additional service-pump/consumer isolations on Sheet005; R20

Sheet 002 | Fire-water storage takeoff

Shown / expected: 1 manual isolation | Status: Present

Confirm sanitary separation/backflow at actual fire-water storage interface in its P&ID.

Sheet 002 | Tank bottom drains

Shown / expected: Individual drain valves for 8 tanks | Status: Not shown

Normally closed drain isolation; retain available overflow; R05

Sheet 003 | Main-pump discharges

Shown / expected: 10 current pump positions: NRV + isolation each | Status: Present

Correct relative sequence and orientation against legend; future symbols excluded from current count.

Sheet 003 | Main-pump suctions

Shown / expected: 10 current pump positions: individual suction isolation | Status: Not shown

Add suitable individual pump suction isolation; R08

Sheet 003 | Suction header

Shown / expected: 2 section valves marked NO | Status: Present

Three sections with both pump groups served; hydraulic/availability proof pending.

Sheet 003 | Discharge headers

Shown / expected: 1 section valve for A and 1 for B, marked NO | Status: Present

Single-end outlets do not prove all isolation cases; R09

Sheet 003 | Group outlets

Shown / expected: FCV-401 / 411; MOV-401 / 411; 2 bypass valves | Status: Present

Bypass NC status and FCV-leg maintenance blocks require clarification; R11

Sheet 003 | FCV maintenance legs

Shown / expected: Separate upstream/downstream FCV-leg blocks | Status: Not shown

Needed if live FCV removal through bypass is intended; R11

Sheet 003 | Water-quality sample taps

Shown / expected: SP-101 on002; SP-401 / 411 on003 with tap valves | Status: Present

Confirm analyser sample-system isolation/conditioning and representative mixing; R13

Sheet 003 | Suction surge

Shown / expected: 1 common isolation for each of 3 typical vessel pairs | Status: Present / incomplete maintenance detail

Individual-vessel isolation/drain/precharge and N-1 surge adequacy; R14

Sheet 003 | Discharge surge

Shown / expected: Each group: common takeoff valve + valve between trains | Status: Present / topology issue

Not independently isolable in both directions; R15

Sheet 003 / 004 | Chemical injection

Shown / expected: Branch injection NRV and isolation at each header | Status: Not shown

Independent dosing and cross-pressure backflow protection; R28

Sheet 003 | Header drains

Shown / expected: 1 drain symbol marked TYP on B discharge | Status: Present / coverage unproven

Other isolated sections and controlled pressure letdown require confirmation; R33

Sheet 004 | NaOCl storage drain

Shown / expected: 1 tank-bottom drain valve | Status: Present / destination issue

Product drain to hyperchlorination route requires compatible collection/treatment; R34

Sheet 004 / 005 | Chemical bund drains

Shown / expected: Controlled, segregated acid/NaOCl drain valves | Status: Not shown

Segregated containment and NC controlled bund release; R30

Sheet 005 | Service-pump discharges

Shown / expected: 1 NRV on each of P-801 / P-802 | Status: Present

Suction/discharge service isolation not shown; R20

Sheet 005 | Cooling / chemical water makeup

Shown / expected: Hazard-appropriate potable backflow protection | Status: Not shown

Select approved separation at each hazardous interface; R21

Sheet 005 | Compressor discharge trains

Shown / expected: Independent NRV and outlet maintenance isolation | Status: Not shown at PFD level

Confirm vendor package devices before common receiver feed; R17

Sheet 005 | Air receivers

Shown / expected: Service isolation, safety relief and condensate drains | Status: Not shown at PFD level

Confirm protection in every permitted isolation state; R18

Sheet 005 | Mobile air source

Shown / expected: Manual isolation at MC-601 | Status: Present / detail incomplete

Cap/blind, NRV and pressure/quality safeguards; R19

Sheet 005 | Cooling loop components

Shown / expected: Local isolation/check/balancing/vent/drain/relief functions | Status: Not shown at PFD level

Confirm actual cooling package/P&ID; R23

Sheet 005 | Sump-pump discharges

Shown / expected: Individual NRV and isolation for P-811 / P-812 | Status: Not shown

Avoid flow back through standby pump; R32

Sheet 005 | Lagoon transfers

Shown / expected: Controlled transfer/release and reverse-flow protection | Status: Not shown

Treatment and approved release philosophy; R35

Source SHA256: 638866ff604c3500a8b0c72dd666cb8e8cc7863817889240f42996c32450021b